A Project Report on

RADAR SYSTEM USING ARDUINO BOARDS

This is submitted in partial fulfilment of

The requirement for the award of the degree of

DIPLOMA

In

Electronics and Communication Engineering

Under the guidance of

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CERTIFICATE

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ABSTRACT

This project is about Radar System controlled via Arduino. This RADAR system consists of an ultrasonic sensor and servo motor. Ultrasonic sensor is attached to the servo motor it rotates about 180 degrees and gives a visual representation of the software called processing IDE. Processing IDE gives graphical representation and it also gives an angle or position of the object and the distance of the object. This system is controlled through Arduino.

Radar (sonar) able to detect the presence of fixed and moved targets and can determine the distance between it and the targets in different angles (0-180)0 which is the range can sweep by Tower Pro SG90 Servo used in this project.

The detection of objects (targets) doesn't depend on the type of the material that the target made from it; it can detect all the types of objects in the range.

It is very important to use these techniques for security in government offices, street, banks, and so and instead of the camera system because it is more accurate

CHAPTER-1
INTRODUCTION

A radar system, short for Radio Detection and Ranging, is a technology that uses radio waves to detect, locate, and track objects in the surrounding environment. It is widely used in various fields, including military, aviation, meteorology, navigation, and traffic control.

The fundamental principle behind radar is the transmission of radio waves, which are electromagnetic waves, from a radar transmitter. These waves travel through the air, and when they encounter an object, they bounce off it and return to a radar receiver. By measuring the time, it takes for the radio waves to travel back and analysing their characteristics, a radar system can determine the distance, direction, speed, and other properties of the objects.

1.1 BASIC CONCEPT OF RADAR

Radar is an electromagnetic system for the detect and determine the locations of objects and determine distance, and ranges. It operates by transmitting a particular type of waveform, a pulse-modulated, and detects the nature of the echo signal. Radar is used to extend the capability of one's senses for observing the environment, especially the sense of vision (Merrill l. Skolnik,1981).

An elementary form of radar consists of a transmitting antenna emitting electromagnetic radiation generated by an oscillator of some

sort, a receiving antenna, and the receiver. A portion of the transmitted signal is intercepted by a reflecting object (target) and is radiated in all directions. The receiving antenna collects the returned energy and delivers it to a receiver.

The distance to the target is determined by measuring the time taken for the radar signal to travel to the target and back. The direction of the target determined from the direction of arrival of the reflected wavefront (Merrill I. Skolnik, 1981).

The basic concept of a radar transmitting a signal an receiving a return from a target is shown in Figure (1-1).

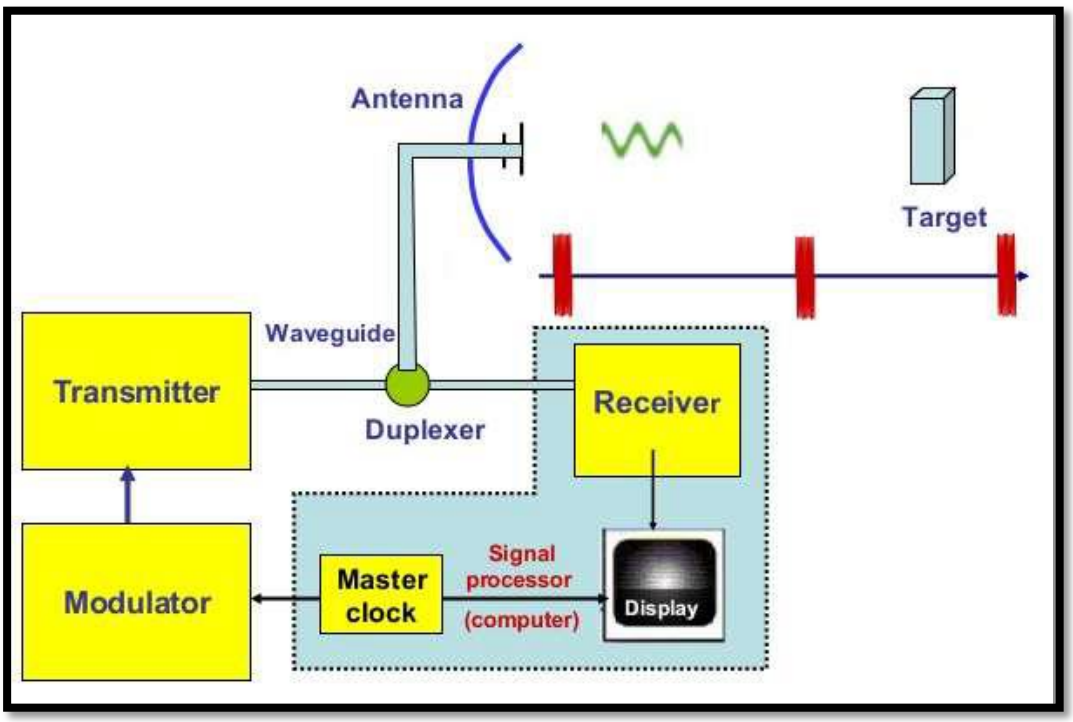

Figure (1.1): Simplified radar block diagram

The radar generates a signal, which is transmitted through an antenna into the desired direction. The antenna is designed to concentrate the radar energy in a particular direction. Only a small

proportion of the transmitted radar energy reaches the target, with the rest missing it or illuminating other nearby objects, including the earth's surface or going off into space, as the radar beam is normally much wider than the angular dimensions of the target. The radar beam broadens as the wave propagates from the radar. So is reduced with the distance the target also reduces with the distance (P. Tait, 2009).

We can observe different objects around the world. Similarly, radar-like radio detection and ranging are used to assist the pilots while flying through fog because the pilot cannot notice that where they are traveling. The radar used in the airplanes is similar to a torchlight that works with radio waves in place of light. The airplane transmits a blinking radar signal and listens out for any indications of that signal from nearby objects. Once the indications are noticed, then the airplane identifies something is near & it uses the time taken for the indications to reach for discovering how distant away it is. This article discusses an overview of Radar and its working.

Similar to several inventions, the radar system is not easy to give credit to an individual because it was the outcome of earlier work on the properties of electromagnetic radiation for the accessibility of numerous electronic devices. The question of main concern is more complicated by the hide of military privacy under which radio location techniques were examined in different countries in the early days of World War-II

1.2 The Echoes Principal

An object (normally referred to as a target) is detected by the transmission of a pulse of radio energy and the subsequent reception of a fraction of such energy (the echo) which is reflected by the target in the direction of the transmitter. The phenomenon is analogous to the reflection of sound waves from land formations. If a blast is sounded on a ship's whistle, the energy travels outward and some of it may strike. The energy which is intercepted will be reflected by the cliff. If the reflected energy returns in the direction of the ship,
and is of sufficient strength, it will be heard as an audible echo which, in duration and tone, resembles the original blast. In considering the echo principle the following points can usefully assist in a preliminary understanding of radar detection:

(a) The echo is never as loud as the original blast.

(b) The chance of detecting an echo depends on the loudness and duration of the original blast.

(c) Short blasts are required if echoes from close targets are not to be *drowned* by the original blast. (d) A sufficiently long interval between blasts is required to allow time for echoes from distant targets to return.

While the sound analogy is extremely useful, it must not be pursued too far, as there are a number of ways in which the character and behaviour of radio waves differ from those of sound waves. In particular at this stage, it is noteworthy that the speed of radio waves is very much higher than that of sound waves.

Chapter-2

Materials & Methods

2.1 MB-102 Solderless Bread Board

The breadboard has 2 split power buses contents from 10 columns, and 63 rows with a total of 830 ties in points. All pins are spaced by a standard 0.1" as shown in figure (2-1).

This board also has a self-adhesive on the back. The boards also.

Plugboard a terminal array board) became available and nowadays the term "breadboard" is commonly used to refer to these. This makes it easy to use for creating temporary prototypes and experimenting with circuit design. A variety of electronic systems may be prototyped by using breadboards, from small analog and digital circuits to complete central processing units (CPUs).

No.	Contents	Characteristics
A-	1 Terminal Strips	Tip-point 630
B-	2 Distribution Strips	Tie-point 200
C-	Solderless breadboard	MB-102
D-	Wire size	20-29 AWG wires.
E-	Size	16.5 x 5.5 x 0.85 cm
Dimensions: **6.5 x 2.1 x 0.38" (165.1 x 54.29 x 9.68mm).**		

Table (2-1): The characteristics of each used part

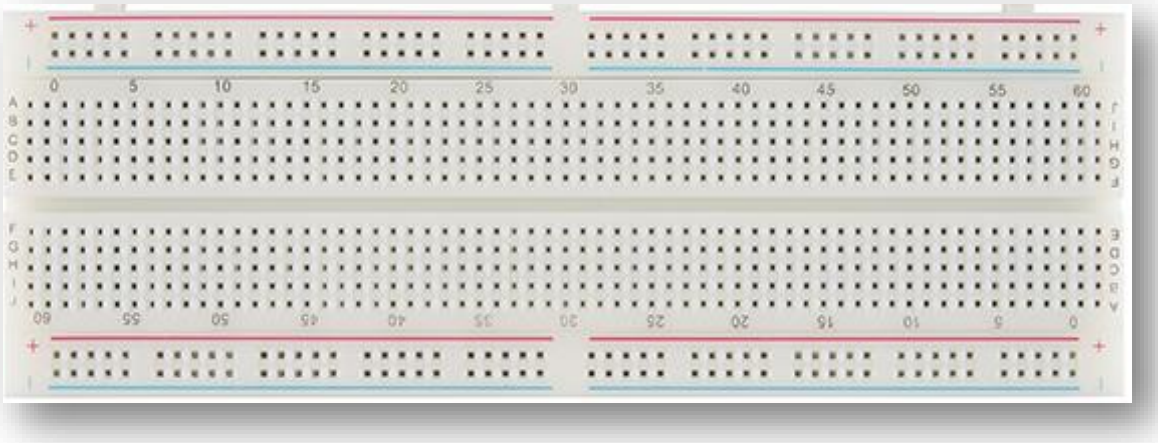

Figure (2.1): Breadboard, 830 Point Solderless PCB Breadboard MB-102.

2.2 JUMPER WIRES

The jumper is an electrical wire, or group of them in a cable, with a pin at each end, which is normally used to interconnect the components of a breadboard internally or with other components

Individual jump wires are fitted by inserting their "end connectors into the slots provided in a breadboard, the header connector of a circuit board or a piece of test component.

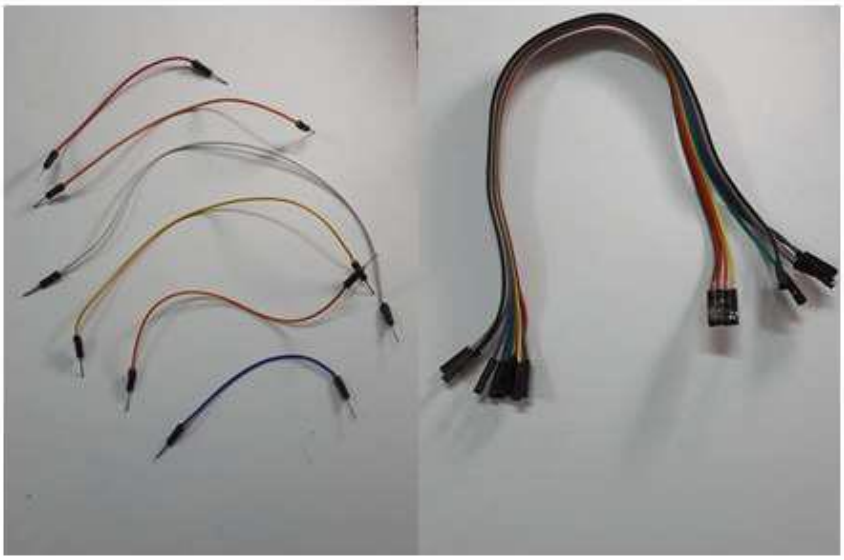

Figure (2.2): Jumper wires with different standards

2.3 Tower PRO TM Microservo SG90

A servomotor is a rotary actuator that allows for precise control of velocity, acceleration, and angular position. It consists of a suitable motor coupled to a sensor for position feedback. It also contains a relatively sophisticated controller for a dedicated module designed specifically for use with servomotors. Servomotors uses servomechanism to achieve closed loop control with a generic open loop motor and are used in applications such as robotics, CNC machinery or automated manufacturing.

SG90 digital servo used to achieve this study, which is the new version of SG90 analog servo and its properties as in the following table (2-2).

Figure (2.3): Tower Pro SG90 Servo

Contents	Characteristics
Modulation:	Analog
Torque:	4.8V: 25.00 oz-in (1.80 kg-cm)
Speed:	4.8V: 0.12 sec/60°
Weight:	9 g
Dimensions:	Length:0.91 in (23.0 mm) Width:0.48 in (12.2 mm) Height:1.14 in (29.0 mm)
Motor Type:	3-pole
Gear Type:	Plastic
Rotation/Support:	Bushing
Pulse Width:	500-2400 µs
Connector Type:	JR

Table (2.3): Tower Pro SG90 Servo Characteristics

2-1-4 The Transceivers

The transceivers work on a principle similar to radar or sonar which evaluate attributes of the objects (targets) by interpreting the echoes from radio or sound waves respectively. Ultrasonic sensors generate high frequency sound waves and evaluate the echo which is received back by the sensor.

The Ultrasonic transmitter transmits an ultrasonic wave, this wave travels in the air and when it gets objected by any material it gets reflected back toward the sensor this reflected wave is observed by the Ultrasonic receiver module.

Figure (2.4): HC-SR04 Sensor

Parameter	Characteristics
Operation Voltage	DC 5V
Oeration Current	15 mA
Operation Frequency	40 Hz
Range	(2 cm – 4m)
Measuring Angle	15 Degree
Trigger Input Signal	10µS TTL pulse
Echo, Output Signal	Input, TTL lever signal and the range in proportion
Dimensions	45 * 20 * 15mm

Table (2-4): HC-SR04 Sensor characteristics

2.5 Arduino UNO

Arduino is an open-source project that created microcontroller-based kits for building digital devices and interactive objects that can sense and control physical devices. In this project the systems provide sets of digital and analog input/output (I/O) pins that can interface to various expansion boards (termed shields) and other circuits. The Arduino project provides an integrated development environment

(IDE) based on a programming language named Processing, which also supports the languages, C and C++ (Margolis, M. 2011).

The Arduino UNO board is probably the most popular among the developers of electronic products based on the Arduino. It is ideal for debugging the software and hardware parts of devices during the development phase (Minns, P.D., 2013).

Parameter	Characteristics
Operation Voltage	DC 5V
Analog input pins	6
DC Current per I/O Pin	40 mA
Clock Speed	16 MHz
Board length	68.6 mm
Board width	53.4 mm
Weight	25 g
Digital I/O Pins 14 6 provide PWM output	

Table 2.5 (Arduino UNO Board Characteristics)

Figure(2.5):Arduino board

Chapter-3
Working Of Radar System

3.1 INTRODUCTION

A processing software and an Arduino software is used with hardware system for detection of objects various parameters. One of the most common applications of ultra-sonic sensor is range finding. It is also called as sonar which is same as radar in which ultrasonic sound is directed at a particular direction and if there is any object in its path it strikes it and gets reflected back and after calculation time taken to come back, we can determine distance of object. in real life this method is used by bats.

In project, we have designed Arduino RADAR Model using Ultrasonic Sensor for Detection & Ranging. RADAR is an object detection system that uses radio waves to identify the range, altitude, direction, and speed of the objects. The radar antenna transmits radio wave pulses that bounce off any object in its path. The object returns a portion of the wave received by the receiver which is in line of sight with the transmitter.

Arduino board sends a signal of +5V to the trig pin of Ultrasonic Sensor HC-SR04 which triggers the sensor. Then it provides rotational action at the servo motor mechanically fitted along with ultrasonic Sensor HC-SR04 so that it can detect the moving objects and locate within 180 degrees.

The figure shown below shows the development life cycle of Radar project which involves various step such as design of different components, their testing, their implementation and implementation of entire system and their testing.

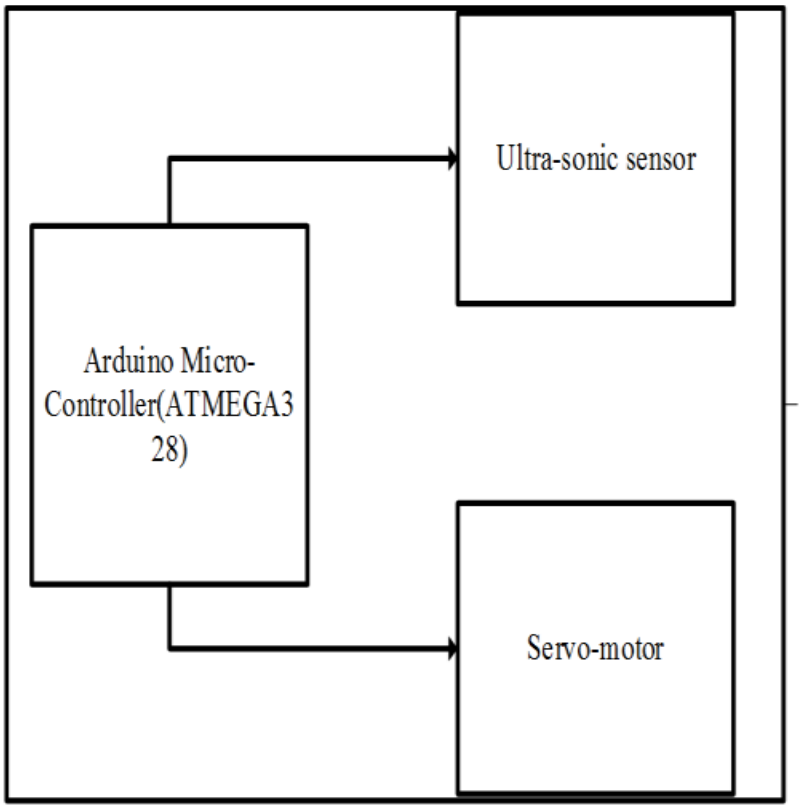

Figure3.1(Hardware System Design of Radar System)

Hardware system consist of basically 3 components named as Arduino, servo-motor, and ultra-sonic sensor. Ultrasonic sensor is mounded upon a servo motor which helps it to move and provide it a turning mechanism. Both ultrasonic sensor and servo motor are controlled and powered by Arduino. As given in above figure 2 we can see both ultrasonic sensor and servo motor is powered by Arduino.

3.2 SYSTEM CIRCUIT DESIGN

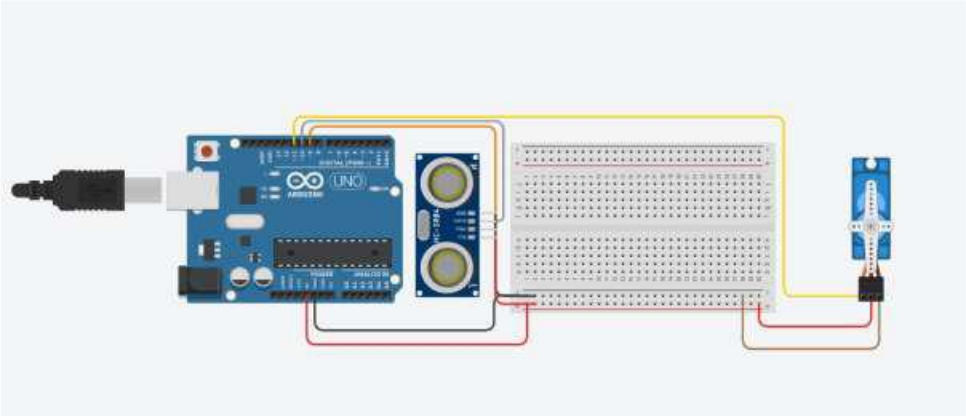

Figure3.2 (System Circuit Design)

Figure shows hardware system design which was designed using fritzing environment. It shows the connection of different electronics components. In the figure triggering pins of ultrasonic sensor is connected to D8 pin of Arduino, control line of servo motor is connected to D6 pin of Arduino and D7 pin of Arduino is connected to echo pin. VCC pins of servo motor and ultrasonic sensor is connected to 5V pin of Arduino while ground pin of Arduino is connected to ground pin of both servo motor and ultra-sonic sensor.

3.3 SYSTEM CIRCUIT IMPLEMENTATION

Implementation of hardware system. It can be seen that ultrasonic servomotor is placed upon a servo motor and it is placed above bread board. Arduino is placed in breadboard in other side of the breadboard and entire connection is made between them.

Arduino and servo motor are stick to breadboard to stop it from tripping over when servo motor moves. Arduino IDE was used to write code and upload it in Arduino. Arduino code reads position of servo motor and calculate distance of nearest object in the path.

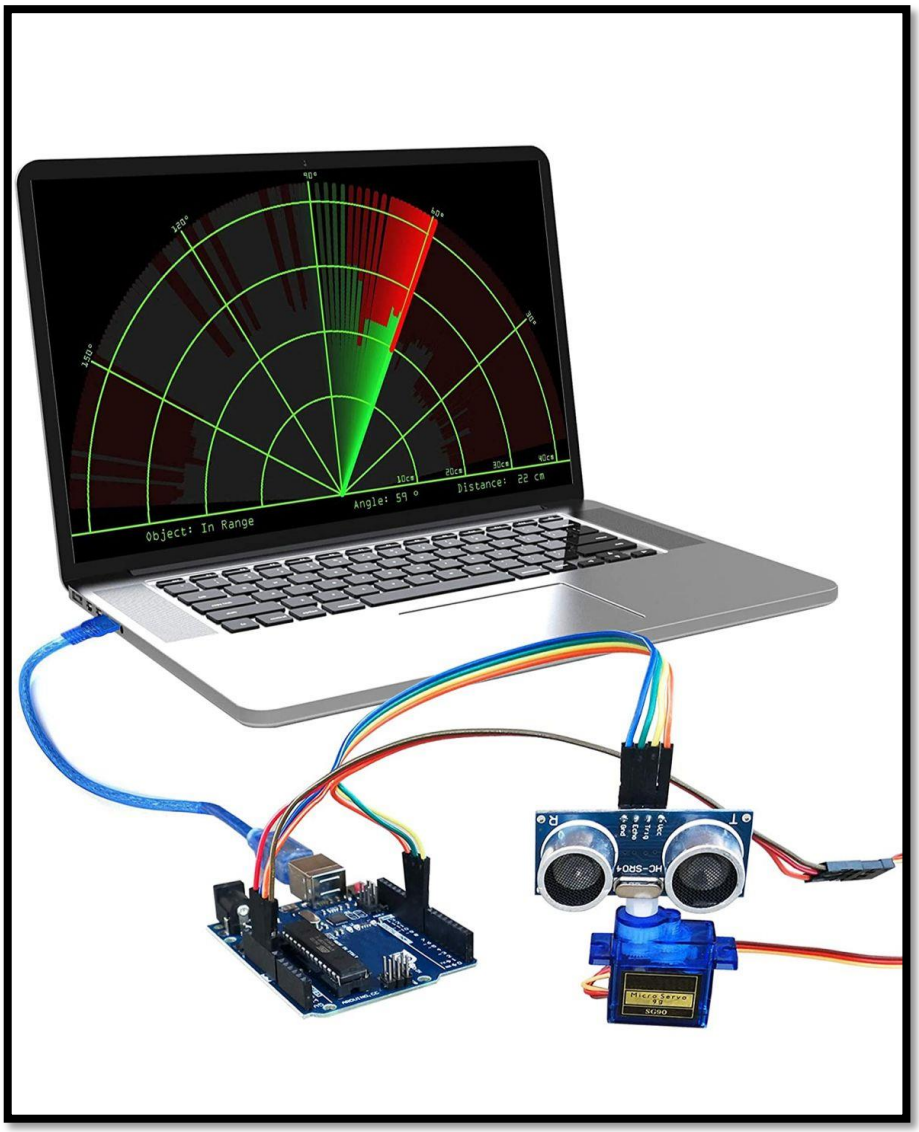

Figure 3.3 (System Circuit Implementation)

CHAPTER-4

RADAR SYSTEM PROGRAMMING CODE

4.1The Arduino for the system

```
// Includes the Servo library
#include <Servo.h>.
// Defines Tirg and Echo pins of the Ultrasonic Sensor
const int trigPin = 10;
const int echoPin = 11;
// Variables for the duration and the distance
long duration;
int distance;
Servo myServo; // Creates a servo object for controlling the servo motor
void setup() {
  pinMode(trigPin, OUTPUT); // Sets the trigPin as an Output
  pinMode(echoPin, INPUT); // Sets the echoPin as an Input
  Serial.begin(9600);
  myServo.attach(12); // Defines on which pin is the servo motor attached
}
```

```
void loop() {
  // rotates the servo motor from 15 to 165 degrees
  for(int i=15;i<=165;i++){
  myServo.write(i);
  delay(30);
  distance = calculateDistance();// Calls a function for calculating the distance measured by the Ultrasonic sensor for each degree

  Serial.print(i); // Sends the current degree into the Serial Port
  Serial.print(","); // Sends addition character right next to the previous value needed later in the Processing IDE for indexing
  Serial.print(distance); // Sends the distance value into the Serial Port
  Serial.print("."); // Sends addition character right next to the previous value needed later in the Processing IDE for indexing
  }
  // Repeats the previous lines from 165 to 15 degrees
  for(int i=165;i>15;i--){
  myServo.write(i);
  delay(30);
  distance = calculateDistance();
  Serial.print(i);
  Serial.print(",");
  Serial.print(distance);
  Serial.print(".");
```

```
  }
}
// Function for calculating the distance measured by the Ultrasonic sensor
int calculateDistance(){


  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  // Sets the trigPin on HIGH state for 10 micro seconds
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  duration = pulseIn(echoPin, HIGH); // Reads the echoPin, returns the sound wave travel time in microseconds
  distance= duration*0.034/2;
  return distance;
}
```

4.2 PROCESSING 4 CODE

This processing code is for mapping interface

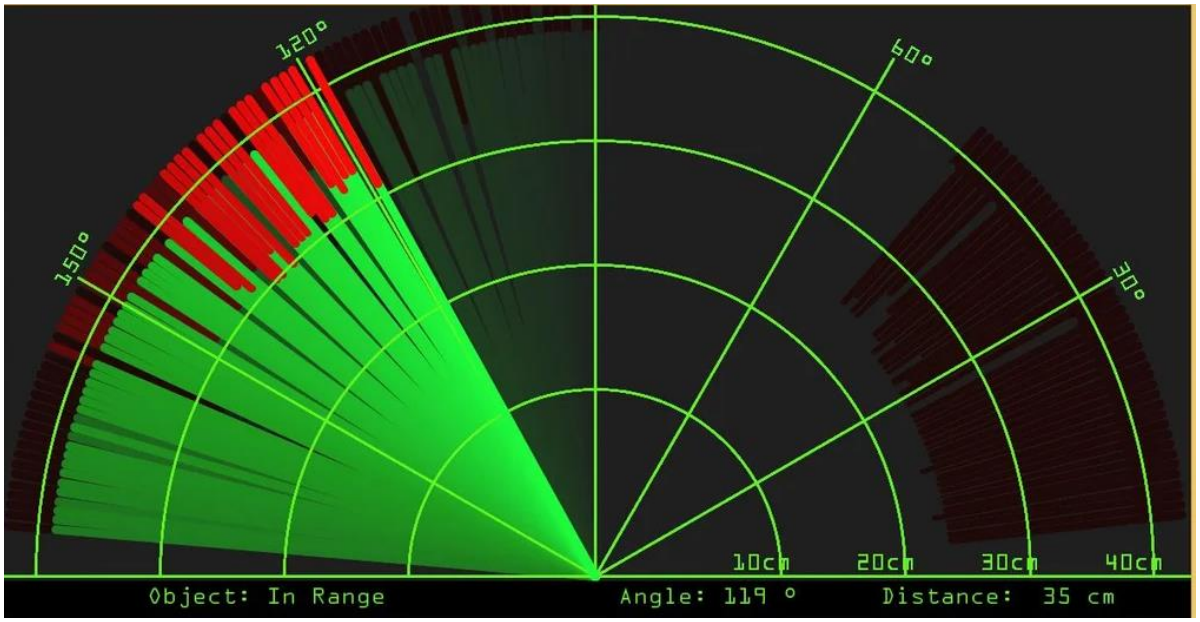

Figure 4.2 GUI Implementation for the mapping interface.

Program code for GUI mapping Interface

```
import processing.serial.*; // imports library for serial communication

import java.awt.event.KeyEvent; // imports library for reading the
data from the serial port

import java.io.IOException;

Serial myPort; // defines Object Serial

// defubes variables

String angle="";

String distance="";

String data="";

String noObject;
```

```
float pixsDistance;
int iAngle, iDistance;
int index1=0;
int index2=0;
PFont orcFont;
void setup() {

 size (1200, 700); // ***CHANGE THIS TO YOUR SCREEN RESOLUTION***
 smooth();
 myPort = new Serial(this,"COM5", 9600); // starts the serial communication
 myPort.bufferUntil('.'); // reads the data from the serial port up to the character '.'. So actually it reads this: angle,distance.
}
void draw() {

  fill(98,245,31);
  // simulating motion blur and slow fade of the moving line
  noStroke();
  fill(0,4);
  rect(0, 0, width, height-height*0.065);

  fill(98,245,31); // green color
```

```
  // calls the functions for drawing the radar
  drawRadar();
  drawLine();
  drawObject();
  drawText();
}
void serialEvent (Serial myPort) { // starts reading data from the
Serial Port
  // reads the data from the Serial Port up to the character '.' and puts it
into the String variable "data".
  data = myPort.readStringUntil('.');
  data = data.substring(0,data.length()-1);

  index1 = data.indexOf(","); // find the character ',' and puts it into the
variable "index1"
  angle= data.substring(0, index1); // read the data from position "0" to
position of the variable index1 or thats the value of the angle the
Arduino Board sent into the Serial Port
  distance= data.substring(index1+1, data.length()); // read the data
from position "index1" to the end of the data pr thats the value of the
distance

  // converts the String variables into Integer
  iAngle = int(angle);
  iDistance = int(distance);
```

```
}
void drawRadar() {
  pushMatrix();
  translate(width/2,height-height*0.074); // moves the starting
coordinats to new location
  noFill();
  strokeWeight(2);
  stroke(98,245,31);
  // draws the arc lines
  arc(0,0,(width-width*0.0625),(width-width*0.0625),PI,TWO_PI);
  arc(0,0,(width-width*0.27),(width-width*0.27),PI,TWO_PI);
  arc(0,0,(width-width*0.479),(width-width*0.479),PI,TWO_PI);
  arc(0,0,(width-width*0.687),(width-width*0.687),PI,TWO_PI);
  // draws the angle lines
  line(-width/2,0,width/2,0);
  line(0,0,(-width/2)*cos(radians(30)),(-width/2)*sin(radians(30)));
  line(0,0,(-width/2)*cos(radians(60)),(-width/2)*sin(radians(60)));
  line(0,0,(-width/2)*cos(radians(90)),(-width/2)*sin(radians(90)));
  line(0,0,(-width/2)*cos(radians(120)),(-width/2)*sin(radians(120)));
  line(0,0,(-width/2)*cos(radians(150)),(-width/2)*sin(radians(150)));
  line((-width/2)*cos(radians(30)),0,width/2,0);
  popMatrix();
}
```

```
void drawObject() {
  pushMatrix();
  translate(width/2,height-height*0.074); // moves the starting
coordinats to new location
  strokeWeight(9);
  stroke(255,10,10); // red color
  pixsDistance = iDistance*((height-height*0.1666)*0.025); // covers
the distance from the sensor from cm to pixels
  // limiting the range to 40 cms
  if(iDistance<40){
    // draws the object according to the angle and the distance
  line(pixsDistance*cos(radians(iAngle)),-
pixsDistance*sin(radians(iAngle)),(width-
width*0.505)*cos(radians(iAngle)),-(width-
width*0.505)*sin(radians(iAngle)));
  }
  popMatrix();
}
void drawLine() {
  pushMatrix();
  strokeWeight(9);
  stroke(30,250,60);
  translate(width/2,height-height*0.074); // moves the starting
coordinats to new location
```

```
  line(0,0,(height-height*0.12)*cos(radians(iAngle)),-(height-
height*0.12)*sin(radians(iAngle))); // draws the line according to the
angle
  popMatrix();
}
void drawText() { // draws the texts on the screen

  pushMatrix();
  if(iDistance>40) {
  noObject = "Out of Range";
  }
  else {
  noObject = "In Range";
  }
  fill(0,0,0);
  noStroke();
  rect(0, height-height*0.0648, width, height);
  fill(98,245,31);
  textSize(25);

  text("10cm",width-width*0.3854,height-height*0.0833);
  text("20cm",width-width*0.281,height-height*0.0833);
  text("30cm",width-width*0.177,height-height*0.0833);
```

```
  text("40cm",width-width*0.0729,height-height*0.0833);
  textSize(40);
  text("Radar System", width-width*0.875, height-height*0.0277);
  text("Angle: " + iAngle +" °", width-width*0.48, height-
height*0.0277);
  text("Distance: ", width-width*0.26, height-height*0.0277);
  if(iDistance<40) {
  text("        " + iDistance +" cm", width-width*0.225, height-
height*0.0277);
  }
  textSize(25);
  fill(98,245,60);
  translate((width-width*0.4994)+width/2*cos(radians(30)),(height-
height*0.0907)-width/2*sin(radians(30)));
  rotate(-radians(-60));
  text("30°",0,0);
  resetMatrix();
  translate((width-width*0.503)+width/2*cos(radians(60)),(height-
height*0.0888)-width/2*sin(radians(60)));
  rotate(-radians(-30));
  text("60°",0,0);
  resetMatrix();
  translate((width-width*0.507)+width/2*cos(radians(90)),(height-
height*0.0833)-width/2*sin(radians(90)));
```

```
  rotate(radians(0));
  text("90°",0,0);
  resetMatrix();
  translate(width-width*0.513+width/2*cos(radians(120)),(height-
height*0.07129)-width/2*sin(radians(120)));
  rotate(radians(-30));
  text("120°",0,0);
  resetMatrix();
  translate((width-width*0.5104)+width/2*cos(radians(150)),(height-
height*0.0574)-width/2*sin(radians(150)));
  rotate(radians(-60));
  text("150°",0,0);
  popMatrix();
}
```

4.3 WORKING

The aim of this project is to calculate the distance position and speed of the object placed at some distance from the sensor. Ultrasonic sensor sends the ultrasonic wave in different directions by rotating with help of servo motor. This wave travels in air and gets reflected back after striking some object. This wave is again sensed by the sensor and its characteristics is analysed and output is displayed in screen showing parameters such as distance and position of object.

Arduino IDE is used to write code and upload coding in Arduino and helps us to sense position of servo motor and posting it to the serial

port along with the distance of the nearest object in its path. The output of sensor is displayed with the help of processing software to give final output in display screen.

4.4 ADVANTAGES

1. Radar procurable value is very low

2. Working and maintenance value is low.

3. Distance active resolution is high

4. Radar's jam is troublesome

5. It can work in any place

6. NASA uses radio detection and ranging to map the world and alternative plants

CONCLUSION

In this paper a system radar system was designed with the help of Arduino, servomotor and ultrasonic sensor which can detect the position,

distance of obstacle which comes in its way and converts it into visually representable form.

This system can be used in robotics for object detection and avoidance system or can also be used for intrusion detection for location sizes.

Range of the system depends upon type of ultra-sonic sensor used. We used HC-SR04 sensor which range from 2 to 40 cm.

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